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Project Title:

The Electronic Gate: The NPN Transistor

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Software Used:

Proteus

Main Component:

2N2222 NPN Transistor

Abstract

This academic report presents the design and simulation of a light-controlled electronic switching circuit using a 2N2222 NPN transistor. The circuit uses an LDR voltage divider to produce a light-dependent control voltage, which is applied to the base of the transistor through a base resistor. The transistor then controls a larger current through an LED load connected to the collector path. The purpose of the circuit is to demonstrate how a small sensor signal can control a larger output current in a safe and measurable way.

The method of the project is based on three connected stages. The first stage is the sensing stage, where a fixed resistor and an LDR form a voltage divider. The second stage is the control stage,

where the divider output is connected to the transistor base through a current-limiting resistor. The third stage is the output stage, where the transistor switches the LED current through the collector-emitter path.

The key calculations verify the SENSE voltage in darkness and bright light, the base current, and the LED current. In darkness, the LDR resistance is high, which makes the SENSE voltage high enough to turn the transistor on. In bright light, the LDR resistance is low, which makes the SENSE voltage too low to activate the transistor. The LED current is limited by a series resistor, and the base current is limited by the base resistor, so the circuit remains electrically safe.

The Proteus simulation is expected to confirm that the LED turns on in darkness and turns off in bright light. The final conclusion is that the 2N2222 transistor can operate as an electronic gate that allows a weak light-dependent signal from an LDR voltage divider to control a larger LED current safely and reliably.

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Chapter 1: Description

1.1 Project Overview

This project is about the design and simulation of a light-controlled transistor switching circuit. The circuit uses an LDR voltage divider to generate a voltage signal that changes according to light intensity. This voltage signal is then used to control the base of a 2N2222 NPN transistor. When the transistor receives sufficient base current, it allows a larger current to flow through the LED load. When the base signal is too low, the transistor stops the LED current.

The project demonstrates an important electronic principle: a small control signal can be used to control a larger output current. The LDR does not directly power the LED. Instead, the LDR creates a control voltage. The transistor receives this control voltage at its base and acts as a controlled switch for the LED current. This makes the circuit more practical than connecting the sensor directly to the output load.

The circuit is designed for simulation in Proteus. Proteus is used to build the schematic, assign component values, observe the LED behavior, and measure the relevant voltages and currents. The simulation allows the circuit to be tested under different light conditions before any physical construction is attempted.

1.2 Aim of the Project

The aim of this project is to design and simulate a light-controlled transistor switch in Proteus using an LDR, a voltage divider, a base resistor, a 2N2222 NPN transistor, an LED, and a current-limiting resistor.

The specific aim is to show that the voltage produced by the LDR divider can control the base of the transistor and that the transistor can switch the LED current on or off depending on the light level. The circuit must also show safe electrical design. This means that the base current must be limited by the base resistor and the LED current must be limited by the LED series resistor.

The project is successful when the circuit satisfies four conditions. First, the SENSE voltage must increase in darkness. Second, the transistor must turn on when the SENSE voltage is high enough. Third, the LED must turn on when the transistor is on. Fourth, the LED must turn off when bright light lowers the SENSE voltage below the transistor activation level.

1.3 Background Theory

The circuit depends on five basic electronic ideas: LDR behavior, voltage divider output, transistor base control, collector-emitter current flow, and switching action.

An LDR is a light-dependent resistor. Its resistance changes according to the amount of light falling on it. In darkness, the resistance of the LDR becomes high. In bright light, the resistance of the LDR becomes low. Because of this property, the LDR can be used as a sensor for light-controlled circuits.

A voltage divider is formed when two resistive components are connected in series across a voltage supply and a voltage is taken from the midpoint between them. In this project, the fixed resistor R1 and the LDR form the voltage divider. The midpoint is called the SENSE node. The voltage at this node changes when the LDR resistance changes.

For this circuit arrangement, R1 is connected between the positive supply and the SENSE node, while the LDR is connected between the SENSE node and ground. In darkness, the LDR

resistance is high, so the SENSE voltage becomes high. In bright light, the LDR resistance is low, so the SENSE voltage becomes low.

The transistor used in this project is a 2N2222 NPN transistor. It has three terminals: base, collector, and emitter. The base is the control terminal. The collector is connected to the output load path. The emitter is connected to ground. When sufficient current enters the base, the transistor allows current to flow from collector to emitter. When there is not enough base current, the transistor blocks the collector current.

The switching action is the main operation of the circuit. In darkness, the high SENSE voltage sends current through the base resistor into the transistor base. This turns the transistor on, and the LED current flows. In bright light, the low SENSE voltage cannot properly activate the base-emitter junction. The transistor remains off, and the LED does not light.

1.4 Main Concept: Transistor as an Electronic Gate

The main concept of the project is that the transistor works as an electronic gate. A mechanical gate controls whether people or objects can pass through an opening. In a similar way, the transistor controls whether electric current can pass through the LED path.

The base of the transistor acts as the control input. The collector-emitter path acts as the controlled output path. When the base receives sufficient current, the transistor opens the path between collector and emitter. This allows current to move from the positive supply, through the LED resistor, through the LED, through the collector-emitter path, and finally to ground.

The important point is that the LDR voltage divider does not need to supply the full LED current. It only needs to provide enough base current to control the transistor. The larger LED current comes from the power supply and is limited by the LED resistor. Therefore, the transistor separates the weak sensor signal from the stronger output current.

In this project, the transistor is not used as a linear audio or signal amplifier. It is used as a switching device. Its purpose is to turn the LED fully on or fully off according to the base signal produced by the LDR voltage divider.

1.5 Functional Behavior of the Circuit

The circuit has two main operating conditions: darkness and bright light.

In darkness, the LDR resistance becomes high. Because the LDR is connected between SENSE and ground, a high LDR resistance makes the SENSE voltage rise. This high voltage is applied through the base resistor to the transistor base. The transistor turns on and allows current to flow through the LED. Therefore, the LED turns on in darkness.

In bright light, the LDR resistance becomes low. A low LDR resistance pulls the SENSE voltage closer to ground. The base of the transistor does not receive enough voltage and current to turn on properly. The transistor stays off, so the LED current is blocked. Therefore, the LED turns off in bright light.

Light Condition LDR Resistance SENSE Voltage Transistor State LED State

Darkness	High	High ON	ON
Bright light	Low	Low OFF	OFF

This functional behavior is the central expected result of the project. If the LED turns on in darkness and turns off in bright light, the circuit behavior is correct for the chosen LDR orientation.

1.6 Required Components

The circuit requires a power source, a sensing stage, a transistor control stage, an output load stage, a ground reference, and measuring instruments. Each component has a specific role, and no essential protection component should be removed.

Component	Value or Type	Function
DC source	5 V	Provides the circuit supply voltage
R1	10 k Ω	Forms the upper part of the voltage divider
LDR	Variable resistance	Detects light by changing resistance
RB	4.7 k Ω	Limits current entering the transistor base
Q1	2N2222	Works as the NPN transistor switch
RLED	470 Ω	Limits current through the LED and transistor
LED	Red LED	Shows the output state of the circuit
Ground	0 V	Provides the common reference node
Voltmeter	DC	Measures voltage in the circuit
Ammeter	DC	Measures current in the circuit

The values are selected so that the circuit is safe and easy to observe in simulation. The 5 V supply is suitable for a basic transistor switching circuit. The 10 k Ω fixed resistor and the LDR create a clear voltage change between dark and bright conditions. The 4.7 k Ω base resistor limits the base current. The 470 Ω LED resistor limits the LED current to a safe value.

1.7 Role of Each Component

1.7.1 DC Source

The DC source provides 5 V to the entire circuit. It supplies energy to both the sensing stage and the LED output stage. The positive terminal of the source is connected to the positive supply rail, and the negative terminal is connected to ground.

The source is necessary because the LDR divider, transistor, and LED all require an electrical supply to operate. Without the DC source, there is no voltage difference, no current path, and no circuit operation.

1.7.2 Fixed Resistor R1

The fixed resistor R1 has a value of 10 kΩ. It works with the LDR to form the voltage divider. R1 is connected between the positive supply and the SENSE node.

The purpose of R1 is to create a controlled voltage at the SENSE node together with the LDR. When the LDR resistance changes, the voltage division between R1 and the LDR also changes. This produces the light-dependent voltage used to control the transistor.

For the selected circuit arrangement, the SENSE voltage is calculated by the voltage divider formula:

$$V_{\text{SENSE}} = V_{\text{CC}} \times R_{\text{LDR}} \div (R1 + R_{\text{LDR}})$$

This formula shows that the SENSE voltage depends directly on the LDR resistance. When the LDR resistance is high, the SENSE voltage is high. When the LDR resistance is low, the SENSE voltage is low.

1.7.3 LDR

The LDR is the light sensor in the circuit. Its resistance changes according to light intensity. In darkness, its resistance becomes high. In bright light, its resistance becomes low.

In this project, the LDR is connected between the SENSE node and ground. This placement is important because it determines the circuit response. With the LDR connected to ground, darkness produces a high SENSE voltage, and bright light produces a low SENSE voltage.

The LDR does not directly control the LED current. Instead, it controls the voltage at the SENSE node. This SENSE voltage then controls the base of the transistor through the base resistor.

1.7.4 Base Resistor RB

The base resistor R_B has a value of $4.7\text{ k}\Omega$. It is connected between the SENSE node and the base of the 2N2222 transistor.

The base resistor is essential because the base-emitter junction of a transistor behaves like a diode. If the SENSE node were connected directly to the base, excessive current could enter the base when the SENSE voltage is high. This could cause incorrect operation, overheating, or damage in a real circuit.

The base resistor limits the current entering the base. In the dark condition, the SENSE voltage is approximately 4.55 V . If the base-emitter voltage is approximately 0.7 V , the base current is calculated as:

$$I_B = (V_{\text{SENSE}} - V_{BE}) \div R_B$$

$$I_B = (4.55\text{ V} - 0.7\text{ V}) \div 4700\ \Omega$$

$$I_B = 3.85\text{ V} \div 4700\ \Omega$$

$$I_B \approx 0.000819\text{ A}$$

$$I_B \approx 0.819\text{ mA}$$

This current is strong enough to switch the transistor on for the selected LED load while still being safely limited.

1.7.5 2N2222 Transistor

The 2N2222 is the main switching component of the circuit. It is an NPN bipolar junction transistor with three terminals: base, collector, and emitter.

In this circuit, the base receives the control signal through R_B . The collector is connected to the LED load. The emitter is connected to ground. This arrangement makes the transistor operate as a low-side switch.

When the base receives sufficient current, the transistor turns on. This allows current to flow through the LED path. When the base current is absent or too small, the transistor turns off and blocks the LED current.

The transistor allows a small base current to control a larger collector current. For reliable switching, the base current should be large enough compared with the collector current. A common switching design check is:

$$I_B \geq I_C \div 10$$

For this project, the collector current is approximately 5.96 mA, so the required base current is approximately:

$$I_{B\text{-required}} = 5.96 \text{ mA} \div 10$$

$$I_{B\text{-required}} \approx 0.596 \text{ mA}$$

The calculated base current is approximately 0.819 mA, which is greater than 0.596 mA. Therefore, the transistor should turn on properly under the dark condition.

1.7.6 LED Resistor RLED

The LED resistor RLED has a value of 470 Ω . It is connected in series with the LED. Its function is to limit the current through the LED and the transistor collector path.

Without RLED, the LED current could become too high. This could damage the LED and overload the transistor. Therefore, RLED is a required safety and operation component.

When the transistor is on, the approximate LED current is calculated as:

$$I_C = (V_{CC} - V_{LED} - V_{CE(sat)}) \div R_{LED}$$

Using the selected values:

$$I_C = (5 \text{ V} - 2.0 \text{ V} - 0.2 \text{ V}) \div 470 \text{ } \Omega$$

$$I_C = 2.8 \text{ V} \div 470 \text{ } \Omega$$

$$I_C \approx 0.00596 \text{ A}$$

$$I_C \approx 5.96 \text{ mA}$$

This current is suitable for an indicator LED in a basic educational switching circuit.

1.7.7 LED

The LED is the output indicator of the circuit. It shows whether the transistor is allowing current to flow through the load path.

When the transistor is on, current flows from the positive supply through RLED, through the LED, through the collector-emitter path of the transistor, and then to ground. The LED lights.

When the transistor is off, the collector-emitter path is blocked. The LED current cannot flow, so the LED remains off.

The LED must be connected with correct polarity. The anode must face the positive supply through RLED, and the cathode must connect toward the transistor collector. If the LED is reversed, it may not light even if the transistor is working correctly.

1.8 Expected Output

The expected output of the circuit is a clear light-controlled switching response. In darkness, the LED should turn on. In bright light, the LED should turn off.

The complete expected behavior is as follows. When the environment is dark, the LDR resistance becomes high. The SENSE voltage rises to approximately 4.55 V. This voltage sends current through the 4.7 k Ω base resistor into the base of the 2N2222 transistor. The transistor turns on, and approximately 5.96 mA flows through the LED path. The LED lights.

When the environment is bright, the LDR resistance becomes low. The SENSE voltage falls to approximately 0.45 V. This voltage is not sufficient to properly turn on the base-emitter junction of the transistor. The transistor remains off, the collector current becomes approximately zero, and the LED stays off.

The transistor must not overheat during correct operation. Safe operation is achieved because the base current is limited by RB and the LED current is limited by RLED. If either of these resistors is removed, the circuit is no longer correctly protected.

The final expected results are:

Condition	Expected SENSE Voltage	Expected Transistor State	Expected LED State
Darkness	Approximately 4.55 V	ON	ON
Bright light	Approximately 0.45 V	OFF	OFF

1.9 Chapter Summary

This chapter introduced the purpose and operation of the light-controlled transistor switching circuit. The project uses an LDR voltage divider to create a control voltage that changes with light intensity. This control voltage is applied through a base resistor to the base of a 2N2222 NPN transistor. The transistor then controls the LED current through its collector-emitter path.

The chapter explained that the circuit has two main states. In darkness, the LDR resistance is high, the SENSE voltage is high, the transistor turns on, and the LED lights. In bright light, the LDR resistance is low, the SENSE voltage is low, the transistor remains off, and the LED does not light.

The chapter also identified the required components and explained their roles. The DC source provides power, R1 and the LDR form the voltage divider, RB limits base current, the 2N2222 transistor acts as the switching device, RLED limits LED current, and the LED displays the output state. The expected output is a safe and clear switching response in which the LED turns on in darkness and turns off in bright light.

Chapter 2

Calculation

2.1 Given Values

This chapter presents the mathematical verification of the light-controlled NPN transistor switching circuit. The purpose of the calculation is to confirm that the LDR voltage divider produces suitable control voltages, that the base current is limited to a safe value, that the LED current is limited to a safe operating range, and that the 2N2222 transistor can operate correctly as a low-side electronic switch.

The circuit is designed with a 5 V direct current supply. The input sensing stage consists of a fixed resistor and an LDR connected as a voltage divider. The output of this divider is called the SENSE node. This SENSE voltage is applied to the base of the transistor through a base resistor. The transistor then controls the current through an LED load connected to the collector side of the circuit.

The numerical values used in the calculation are shown below.

Quantity	Symbol	Value
Supply voltage	VCC	5 V
Fixed resistor	R1	10 k Ω
LDR resistance in darkness	RLDR dark	100 k Ω
LDR resistance in brightness	RLDR bright	1 k Ω
Base resistor	RB	4.7 k Ω
LED resistor	RLED	470 Ω
LED forward voltage	VLED	2.0 V
Transistor saturation voltage	VCE sat	0.2 V

Quantity	Symbol	Value
Base-emitter voltage	V_{BE}	0.7 V

The selected values are suitable for an educational transistor switching circuit because the sensor section, base section, and output section are each controlled by a resistor. The fixed resistor and LDR determine the SENSE voltage. The base resistor limits the current entering the base of the transistor. The LED resistor limits the current flowing through the LED and the collector-emitter path of the transistor. Therefore, no important current path is left uncontrolled.

2.2 Voltage Divider Formula

The LDR and the fixed resistor form a voltage divider. In this circuit, the fixed resistor R1 is connected between the 5 V supply and the SENSE node, while the LDR is connected between the SENSE node and ground. Because of this arrangement, the SENSE voltage depends directly on the resistance of the LDR.

The voltage divider equation is:

$$V_{SENSE} = V_{CC} \times R_{LDR} \div (R1 + R_{LDR})$$

Where:

V_{SENSE} is the output voltage at the SENSE node.

V_{CC} is the supply voltage.

R_{LDR} is the resistance of the LDR under the present light condition.

R1 is the fixed resistor connected between the supply and the SENSE node.

This formula shows that when the LDR resistance increases, the SENSE voltage also increases. When the LDR resistance decreases, the SENSE voltage also decreases. Therefore, this particular arrangement produces a high SENSE voltage in darkness and a low SENSE voltage in bright light.

This behavior is essential for the selected circuit because the transistor is intended to switch ON in darkness and switch OFF in bright light.

2.3 Dark Condition Calculation

In darkness, the LDR resistance is assumed to be high. For this design, the dark resistance of the LDR is taken as:

$$R_{LDR} = 100 \text{ k}\Omega$$

Using the voltage divider formula:

$$V_{SENSE} = V_{CC} \times R_{LDR} \div (R_1 + R_{LDR})$$

Substitute the known values:

$$V_{SENSE} = 5 \times 100 \text{ k}\Omega \div (10 \text{ k}\Omega + 100 \text{ k}\Omega)$$

First, add the two resistances in the denominator:

$$10 \text{ k}\Omega + 100 \text{ k}\Omega = 110 \text{ k}\Omega$$

Therefore:

$$V_{SENSE} = 5 \times 100 \text{ k}\Omega \div 110 \text{ k}\Omega$$

The resistance unit cancels because the numerator and denominator both contain kilo-ohms:

$$V_{SENSE} = 500 \div 110$$

$$V_{SENSE} \approx 4.55 \text{ V}$$

The calculated SENSE voltage in darkness is approximately:

$$V_{SENSE \text{ dark}} \approx 4.55 \text{ V}$$

This value is significantly greater than the approximate base-emitter turn-on voltage of 0.7 V. Therefore, in darkness, the transistor base receives a sufficiently high control voltage through the base resistor. This allows base current to flow from the SENSE node through the base resistor into the base-emitter junction of the transistor.

The conclusion for the dark condition is that the input stage produces a strong control signal. As a result, the transistor should turn ON, allowing current to flow through the LED load.

2.4 Bright Condition Calculation

In bright light, the LDR resistance is assumed to be low. For this design, the bright resistance of the LDR is taken as:

$$R_{LDR} = 1 \text{ k}\Omega$$

Using the voltage divider formula:

$$V_{SENSE} = V_{CC} \times R_{LDR} \div (R_1 + R_{LDR})$$

Substitute the known values:

$$V_{SENSE} = 5 \times 1 \text{ k}\Omega \div (10 \text{ k}\Omega + 1 \text{ k}\Omega)$$

First, add the two resistances in the denominator:

$$10 \text{ k}\Omega + 1 \text{ k}\Omega = 11 \text{ k}\Omega$$

Therefore:

$$V_{SENSE} = 5 \times 1 \text{ k}\Omega \div 11 \text{ k}\Omega$$

The resistance unit cancels:

$$V_{SENSE} = 5 \div 11$$

$$V_{SENSE} \approx 0.45 \text{ V}$$

The calculated SENSE voltage in bright light is approximately:

$$V_{SENSE \text{ bright}} \approx 0.45 \text{ V}$$

This value is below the approximate base-emitter turn-on voltage of 0.7 V. Therefore, in bright light, the base-emitter junction is not sufficiently forward biased. Since the base does not receive enough voltage to conduct properly, the base current becomes approximately zero.

The conclusion for the bright condition is that the input stage produces a low control signal. As a result, the transistor should remain OFF, and the LED should not conduct current.

2.5 Base Current Calculation

The base current must be calculated to verify that the transistor receives enough drive current in darkness while remaining protected from excessive current. The base resistor R_B is placed between the SENSE node and the base of the transistor. Its function is to limit the current entering the base-emitter junction.

The base current is calculated using:

$$I_B = (V_{SENSE} - V_{BE}) \div R_B$$

Where:

I_B is the base current.

V_{SENSE} is the voltage at the SENSE node.

V_{BE} is the approximate base-emitter voltage of the transistor.

R_B is the base resistor.

In the dark condition:

$$V_{SENSE} \approx 4.55 \text{ V}$$

The approximate base-emitter voltage is:

$$V_{BE} \approx 0.7 \text{ V}$$

The base resistor is:

$$R_B = 4.7 \text{ k}\Omega = 4700 \text{ }\Omega$$

Substitute the values into the base current equation:

$$I_B = (4.55 - 0.7) \div 4700$$

Subtract the base-emitter voltage from the SENSE voltage:

$$4.55 - 0.7 = 3.85 \text{ V}$$

Therefore:

$$I_B = 3.85 \div 4700$$

$$I_B \approx 0.000819 \text{ A}$$

Convert amperes to milliamperes:

$$0.000819 \text{ A} = 0.819 \text{ mA}$$

Thus, the base current in darkness is approximately:

$$I_{B \text{ dark}} \approx 0.819 \text{ mA}$$

This value is suitable for the circuit because it provides enough current to drive the transistor while remaining limited by the base resistor. The base resistor therefore performs two functions at the same time: it allows transistor switching and prevents uncontrolled base current.

2.6 LED Current Calculation

The LED current is the same as the collector current when the transistor is ON because the LED, LED resistor, collector, and emitter are connected in one series current path. The LED current must be calculated to verify that the output load current remains controlled.

When the transistor is saturated, the LED current is calculated using:

$$I_C = (V_{CC} - V_{LED} - V_{CE \text{ sat}}) \div R_{LED}$$

Where:

I_C is the collector current.

V_{CC} is the supply voltage.

V_{LED} is the forward voltage of the LED.

$V_{CE \text{ sat}}$ is the collector-emitter saturation voltage of the transistor.

R_{LED} is the LED current-limiting resistor.

Substitute the given values:

$$I_C = (5 - 2.0 - 0.2) \div 470$$

First, subtract the LED forward voltage and transistor saturation voltage from the supply voltage:

$$5 - 2.0 - 0.2 = 2.8 \text{ V}$$

Therefore:

$$I_C = 2.8 \div 470$$

$$I_C \approx 0.00596 \text{ A}$$

Convert amperes to milliamperes:

$$0.00596 \text{ A} = 5.96 \text{ mA}$$

Thus, the collector current and LED current are approximately:

$$I_C \approx 5.96 \text{ mA}$$

This result shows that the LED current is limited by the $470 \, \Omega$ resistor. The circuit therefore does not allow the LED or transistor collector path to draw uncontrolled current. The LED should illuminate clearly while remaining within a conservative operating range for a small indicator LED.

2.7 Transistor Saturation Check

The transistor must be checked for reliable switching. In a switching circuit, it is not enough for the transistor to conduct slightly. The transistor should enter saturation when it is intended to be fully ON. Saturation means that the transistor behaves like a closed electronic switch, with a small voltage drop between collector and emitter.

A practical forced switching rule is:

$$I_B \geq I_C \div 10$$

This rule means that the base current should be at least one tenth of the collector current. This does not depend on using the maximum current gain of the transistor. Instead, it gives the transistor a strong base drive so that it can turn ON reliably.

The calculated collector current is:

$$I_C \approx 5.96 \text{ mA}$$

The required base current is therefore:

$$I_{B \text{ required}} = I_C \div 10$$

Substitute the collector current:

$$I_{B \text{ required}} = 5.96 \text{ mA} \div 10$$

$$I_{B \text{ required}} \approx 0.596 \text{ mA}$$

From the base current calculation, the actual base current in darkness is:

$$I_{B \text{ actual}} \approx 0.819 \text{ mA}$$

Now compare the actual base current with the required base current:

$$0.819 \text{ mA} > 0.596 \text{ mA}$$

The actual base current is greater than the minimum required base current for forced switching. Therefore, the transistor should enter saturation in darkness.

This confirms that the circuit is not merely producing a high base voltage. It is also producing enough base current to switch the LED load correctly. The transistor should therefore operate as a reliable electronic switch in the dark condition.

2.8 Bright-State OFF Verification

The bright-state condition must also be verified. A correct switching circuit must not only turn ON in the intended condition. It must also turn OFF in the opposite condition.

From the bright-condition voltage divider calculation:

$$V_{\text{SENSE bright}} \approx 0.45 \text{ V}$$

The approximate voltage needed to forward bias the base-emitter junction is:

$$V_{\text{BE}} \approx 0.7 \text{ V}$$

Since:

$$0.45 \text{ V} < 0.7 \text{ V}$$

The base-emitter junction is not sufficiently forward biased in bright light. Therefore, the base current becomes approximately zero:

$$I_{\text{B bright}} \approx 0 \text{ mA}$$

If the base current is approximately zero, the transistor cannot support collector current through normal switching action. Therefore:

$$I_{\text{C bright}} \approx 0 \text{ mA}$$

As a result, the LED remains OFF in bright light.

This verifies that the same circuit has two stable and predictable operating states. In darkness, the SENSE voltage is high enough to turn the transistor ON. In bright light, the SENSE voltage is too low to turn the transistor ON. The mathematical behavior therefore matches the intended circuit operation.

2.9 Power and Safety Check

The safety check verifies that the circuit does not depend on uncontrolled current flow. A correct transistor switching circuit must limit the current in both the base path and the load path. In this circuit, the base current is limited by R_B , while the LED current is limited by R_{LED} .

2.9.1 LED Resistor Safety

The LED resistor is:

$$R_{LED} = 470 \, \Omega$$

The calculated LED current is:

$$I_C \approx 5.96 \, \text{mA}$$

This current is moderate for an indicator LED circuit. The resistor prevents the LED from being connected directly across the supply through the transistor. Without this resistor, the LED current would be determined mainly by the supply, the LED, the transistor, and wiring resistance, which would be unsafe and unsuitable for a controlled circuit.

Therefore, the LED resistor is essential because it defines the output current and protects both the LED and the collector-emitter path of the transistor.

2.9.2 Base Resistor Safety

The base resistor is:

$$R_B = 4.7 \, \text{k}\Omega$$

The calculated base current in darkness is:

$$I_B \approx 0.819 \, \text{mA}$$

This value is large enough to drive the transistor into saturation for the selected LED current, but it is still limited by the resistor. Without the base resistor, the base-emitter junction could receive excessive current when the SENSE voltage is high. This could cause incorrect simulation behavior and could damage the transistor in a physical circuit.

Therefore, the base resistor is essential because it protects the transistor input while still allowing the base to receive enough current for switching.

2.9.3 Transistor Safety

The collector current is approximately:

$$I_C \approx 5.96 \text{ mA}$$

This is a small switching current for a 2N2222 transistor in an educational LED circuit. The transistor is not being asked to drive a large motor, relay coil, lamp, or high-current load. It is only controlling a small LED current that is already limited by RLED.

The base current is also controlled:

$$I_B \approx 0.819 \text{ mA}$$

Therefore, both the input side and output side of the transistor are resistor-limited. This confirms that the transistor is operating under controlled switching conditions.

The circuit is electrically safe under the stated design assumptions because the base path has RB, the LED path has RLED, the emitter is grounded, and the collector current remains low.

2.10 Calculation Summary

The calculations confirm that the circuit works as a light-controlled transistor switch. The voltage divider produces a high SENSE voltage in darkness and a low SENSE voltage in bright light. The high dark-state SENSE voltage creates enough base current to saturate the 2N2222 transistor. The low bright-state SENSE voltage is not enough to forward bias the base-emitter junction, so the transistor remains OFF.

The LED current is limited by the 470 Ω resistor to approximately 5.96 mA. The base current is limited by the 4.7 kΩ resistor to approximately 0.819 mA in darkness. The actual base current is greater than the required forced switching current of approximately 0.596 mA, so the transistor should switch ON reliably in darkness.

The final calculated results are shown below.

Condition	VSENSE	IB	IC	Transistor State	LED State
Dark	4.55 V	0.819 mA	5.96 mA	ON	ON
Bright	0.45 V	approximately 0 mA	approximately 0 mA	OFF	OFF

The calculation chapter proves that the selected component values are coherent. The LDR divider provides the correct control voltage, the base resistor provides controlled base current, the LED resistor provides controlled load current, and the transistor operates as a reliable low-

side switch. Therefore, the mathematical design supports the expected Proteus simulation result: the LED turns ON in darkness and turns OFF in bright light.

Chapter 3

Scheme

3.1 Purpose of the Scheme

This chapter presents the exact circuit scheme used for the light-controlled transistor switching system. The purpose of the scheme is to define the physical and electrical arrangement of all components in Proteus so that the circuit can be constructed, simulated, checked, and explained without ambiguity. The chapter identifies each component, states its position in the circuit, describes the connection of every wire, and explains the current paths that allow the circuit to operate correctly.

The scheme is based on a 5 V direct current supply, an LDR voltage divider, a 2N2222 NPN transistor, a base resistor, an LED load, and a current-limiting resistor for the LED. The LDR and the fixed resistor create a variable voltage at the SENSE node. This voltage is applied to the base of the transistor through the base resistor. When the SENSE voltage is sufficiently high, base current flows, the transistor turns on, and current passes through the LED load. When the SENSE voltage is low, the base current is not sufficient to turn the transistor on, and the LED remains off.

The scheme is therefore not only a drawing of components. It is the exact electrical structure that proves how the sensing stage, the control stage, and the output stage are connected into one working circuit. Every connection must be correct because a single wrong wire can change the behaviour of the transistor, reverse the LED state, remove current limitation, or prevent the simulation from working.

3.2 Proteus Components Used

The Proteus schematic uses a fixed set of components. Each component has a specific electrical function and must be selected carefully from the Proteus component library.

The power source is selected as a direct current source or battery and is set to 5 V. This source supplies energy to both the voltage divider and the LED load. A ground terminal is also required because every measured voltage in the circuit must have a common reference point.

The fixed resistor R1 is selected as a standard resistor and is assigned a value of 10 k Ω . This resistor forms the upper part of the voltage divider. The LDR is selected as the light-dependent resistor component and forms the lower part of the voltage divider. The midpoint between R1 and the LDR is named the SENSE node.

The transistor is selected as a 2N2222 or 2N2222A NPN transistor. This transistor is used as an electronic switch. The base of the transistor receives the control signal from the SENSE node through the base resistor RB. The collector is connected to the LED load. The emitter is connected to ground.

The base resistor RB is selected as a standard resistor and is assigned a value of 4.7 k Ω . Its purpose is to limit the current entering the base of the transistor. The LED resistor RLED is selected as a standard resistor and is assigned a value of 470 Ω . Its purpose is to limit the current through the LED and the transistor collector path.

The output indicator is a red LED. The LED is connected in series with RLED and the collector of the transistor. A direct current voltmeter is used to measure the voltage at the SENSE node with respect to ground. A direct current ammeter is used to measure the current flowing through the LED load path.

The components required in Proteus are therefore: direct current source or battery, resistor, LDR, 2N2222 or 2N2222A transistor, red LED, ground terminal, direct current voltmeter, and direct current ammeter.

3.3 Circuit Block Structure

The scheme is divided into three main functional blocks. This division makes the operation of the circuit easier to understand and also makes the troubleshooting process more systematic. The three blocks are the sensor block, the control block, and the output block.

The sensor block detects the light condition and converts it into a voltage. The control block receives this voltage and controls the transistor base. The output block contains the LED load and shows the final result of the transistor switching action.

This block structure is important because the LDR does not directly power the LED. The LDR creates only a control voltage. The transistor receives this control voltage and switches the LED current. The LED current comes from the 5 V source, not from the LDR itself.

3.3.1 Sensor Block

The sensor block contains R1, the LDR, and the SENSE node. R1 is connected between the positive 5 V rail and the SENSE node. The LDR is connected between the SENSE node and ground. These two components form a voltage divider.

The SENSE node is the most important node in the sensor block because it carries the light-dependent voltage that controls the transistor. When the LDR resistance is high, the voltage at SENSE becomes high. When the LDR resistance is low, the voltage at SENSE becomes low.

The voltage at the SENSE node is determined by the voltage divider relationship:

$$V_{\text{SENSE}} = V_{\text{CC}} \times R_{\text{LDR}} \div (R1 + R_{\text{LDR}})$$

In this scheme, R1 is placed above the SENSE node and the LDR is placed below the SENSE node. Because of this arrangement, darkness increases the SENSE voltage and bright light decreases the SENSE voltage. This means the chosen scheme is designed for the LED to turn on in darkness and turn off in bright light.

The sensor block therefore performs the first essential function of the circuit. It converts light intensity into an electrical voltage that can be used by the transistor control stage.

3.3.2 Control Block

The control block contains the base resistor RB and the base emitter junction of the 2N2222 transistor. The SENSE node is connected to one side of RB, and the other side of RB is connected to the base of the transistor.

RB is necessary because the base emitter junction behaves like a forward-biased semiconductor junction when the base voltage becomes high enough. If the base were connected directly to the SENSE node, excessive base current could flow. This could make the transistor heat up or behave incorrectly in simulation and in a real circuit. RB prevents this by limiting the current entering the base.

When the SENSE voltage is greater than the approximate base emitter turn-on voltage, current flows through RB into the base. The approximate base current can be expressed as:

$$I_B \approx (V_{\text{SENSE}} - V_{\text{BE}}) \div R_B$$

For this circuit, the base emitter voltage is approximately 0.7 V when the transistor is conducting. Therefore, the base current becomes significant only when the SENSE voltage rises above this value.

The control block is the decision stage of the circuit. It decides whether the transistor remains off or turns on. In darkness, the SENSE voltage is high enough to allow base current. In bright light, the SENSE voltage is too low to produce sufficient base current.

3.3.3 Output Block

The output block contains RLED, the red LED, and the collector emitter path of the 2N2222 transistor. RLED is connected from the positive 5 V rail to the LED anode. The LED cathode is connected to the collector of the transistor. The emitter of the transistor is connected to ground.

This arrangement makes the 2N2222 transistor operate as a low side switch. The transistor does not supply current to the LED from its base. Instead, it completes or interrupts the current path from the 5 V supply through RLED, through the LED, through the collector emitter path, and then to ground.

When the transistor is on, the collector emitter path conducts and the LED current flows. When the transistor is off, the collector emitter path is open, so the LED current cannot flow.

The approximate LED current during the on state is calculated by:

$$I_C \approx (V_{CC} - V_{LED} - V_{CE(sat)}) \div R_{LED}$$

For this project, the LED current is limited by the 470 Ω resistor. This ensures that the LED and transistor remain within a safe operating range. The output block therefore provides the visible result of the circuit while keeping the load current controlled.

3.4 Exact Wire Connections

The exact wire connections define the circuit without uncertainty. Each wire has a specific purpose. A correct scheme requires that the voltage divider, transistor, LED, and measuring instruments all share the same electrical reference.

3.4.1 Power Connections

The positive terminal of the 5 V direct current source is connected to the positive supply rail of the circuit. This rail is used to supply both the voltage divider and the LED load. The negative terminal of the 5 V source is connected to the ground rail. The ground rail is the common reference point for the voltage divider, the transistor emitter, and the measuring instruments.

The circuit must not be simulated without a ground reference. Without ground, the node voltages are undefined, and Proteus may produce incorrect results or may fail to simulate the circuit correctly.

The required power connections are as follows. The positive terminal of V1 is connected to the 5 V rail. The negative terminal of V1 is connected to ground.

3.4.2 LDR Voltage Divider Connections

The LDR voltage divider is connected so that the fixed resistor is above the SENSE node and the LDR is below the SENSE node. This exact placement is essential because reversing the two components changes the output behaviour of the circuit.

The 5 V rail is connected to the first terminal of R1. The second terminal of R1 is connected to the SENSE node. The SENSE node is connected to the first terminal of the LDR. The second terminal of the LDR is connected to ground.

With this arrangement, the divider current flows from the 5 V rail through R1, then through the SENSE node, then through the LDR, and finally to ground. The SENSE node voltage depends on the resistance of the LDR. In darkness, the LDR resistance becomes large, so the SENSE voltage becomes high. In bright light, the LDR resistance becomes small, so the SENSE voltage becomes low.

This connection is therefore responsible for producing the correct control voltage for the transistor.

3.4.3 Base Control Connections

The SENSE node is connected to one terminal of the base resistor RB. The other terminal of RB is connected to the base of Q1. This connection transfers the light-dependent voltage from the sensor block to the transistor control input.

The base must not be connected directly to the SENSE node. The base resistor must remain between the SENSE node and the base terminal. This is necessary because the base emitter

junction requires current limitation. Without RB, the circuit would lose safe control over the base current.

The base control connection therefore has one correct form: SENSE node to RB, and RB to the base of Q1. Any direct connection from SENSE to the base without RB is incorrect.

3.4.4 Transistor Output Connections

The emitter of Q1 is connected directly to ground. This connection gives the transistor a fixed reference point and allows base current and collector current to return to the supply source through the ground path.

The collector of Q1 is connected to the cathode side of the LED load path. This placement allows Q1 to control whether the LED current can reach ground. When Q1 is on, the collector emitter path conducts and the LED turns on. When Q1 is off, the collector emitter path does not conduct and the LED turns off.

The transistor output connection must follow this rule: emitter to ground and collector to the LED cathode side. If the collector and emitter are swapped, the transistor may not switch correctly, the simulation readings may become unreliable, and the LED may fail to operate as expected.

3.4.5 LED Load Connections

The LED load is connected in series with its current-limiting resistor. The 5 V rail is connected to the first terminal of RLED. The second terminal of RLED is connected to the anode of the LED. The cathode of the LED is connected to the collector of Q1.

This connection ensures that current flows in the correct direction through the LED when the transistor is on. The LED anode must face the positive supply side, and the LED cathode must face the transistor collector. If the LED is reversed, it will not conduct under normal operating conditions, and the output indication will be incorrect.

RLED must remain in series with the LED. It is not optional. Its function is to limit the LED current and also protect the transistor collector path. Without RLED, the LED current could become excessive.

The correct LED load path is therefore: 5 V rail to RLED, RLED to LED anode, LED cathode to Q1 collector, Q1 emitter to ground.

3.4.6 Voltmeter Connections

The direct current voltmeter is connected to measure the SENSE voltage. The positive terminal of the voltmeter is connected to the SENSE node, and the negative terminal of the voltmeter is connected to ground.

The voltmeter must be connected in parallel with the measured node and the reference point. It must not be placed in series with the current path. Its purpose is to measure potential difference, not current.

This measurement is essential because the SENSE voltage determines whether the transistor should turn on or remain off. In the dark condition, the voltmeter should show a high SENSE voltage. In the bright condition, the voltmeter should show a low SENSE voltage.

For this circuit, the expected values are approximately 4.55 V in darkness and approximately 0.45 V in bright light, based on the selected LDR resistance values.

3.4.7 Ammeter Connections

The direct current ammeter is used to measure the LED load current. It must be placed in series with the LED current path. A correct placement is between the LED cathode and the collector of Q1.

In this placement, the LED cathode is connected to the positive terminal of the ammeter, and the negative terminal of the ammeter is connected to the collector of Q1. This allows the ammeter to measure the same current that flows through the LED and the collector emitter path.

The ammeter must not be connected in parallel with the LED, the resistor, or the supply source. A parallel ammeter connection can create an unintended low-resistance path and may cause incorrect current readings or simulation errors.

The expected ammeter reading when the transistor is fully on is approximately 5.96 mA. When the transistor is off, the ammeter should show approximately zero current.

3.5 Correct Current Paths

The circuit has three essential current paths. These paths must be understood separately because each one performs a different function. The divider current produces the sensor voltage. The base current controls the transistor. The LED current produces the output.

3.5.1 Divider Current Path

The divider current path begins at the positive 5 V rail. Current flows through R1, reaches the SENSE node, continues through the LDR, and returns to ground.

The path is:

+5 V → R1 → SENSE → LDR → GND

This current path creates the SENSE voltage. The value of the SENSE voltage depends on the ratio between R1 and the LDR resistance. If the LDR resistance changes, the voltage division changes, and the SENSE voltage changes.

This path must exist before the transistor can be controlled. If R1 is disconnected, if the LDR is disconnected, or if the ground connection is missing, the SENSE node may become floating or incorrect.

3.5.2 Base Current Path

The base current path begins at the SENSE node when the SENSE voltage is high enough. Current flows from SENSE through RB, enters the base of the transistor, passes through the base emitter junction, and returns to ground through the emitter.

The path is:

SENSE → RB → Base → Emitter → GND

This path allows the SENSE voltage to control the transistor. The base current does not directly power the LED. Its function is to activate the transistor so that the collector current path can conduct.

The base resistor is the controlling boundary of this path. It limits the current entering the base and prevents the base emitter junction from being overdriven.

3.5.3 LED Current Path

The LED current path begins at the positive 5 V rail. Current flows through RLED, then through the LED from anode to cathode, then into the collector of Q1, then through the collector emitter path, and finally to ground.

The path is:

+5 V → RLED → LED → Collector → Emitter → GND

This current path exists only when the transistor is on. If the transistor is off, the collector emitter path does not conduct, so the LED current is interrupted.

The LED current is limited by RLED. This makes the output stage safe and predictable. The approximate current in the on state is:

$$I_C \approx (V_{CC} - V_{LED} - V_{CE(sat)}) \div R_{LED}$$

Using the selected values:

$$I_C \approx (5 - 2.0 - 0.2) \div 470$$

$$I_C \approx 2.8 \div 470$$

$$I_C \approx 0.00596 \text{ A}$$

$$I_C \approx 5.96 \text{ mA}$$

This current is suitable for a basic LED indicator circuit.

3.6 Correct Transistor Pin Use

The 2N2222 transistor has three functional terminals: base, collector, and emitter. The base is the control terminal, the collector is the load-side terminal, and the emitter is the ground-side terminal in this circuit.

The base receives the control signal only through RB. This is necessary because the base current must be limited. The collector connects to the LED cathode side, which makes the transistor responsible for completing the LED current path. The emitter connects directly to ground, which gives the transistor a fixed reference and provides the return path for both base current and collector current.

Correct pin use is essential because the transistor is not symmetrical. The base, collector, and emitter cannot be exchanged without changing the behaviour of the circuit. If the emitter is not grounded, the transistor may not receive the correct base emitter voltage. If the collector is not connected to the LED load, the transistor cannot control the output current. If the base is connected incorrectly, the transistor cannot be controlled by the LDR voltage divider.

In Proteus, the pin labels of the transistor should always be checked directly. The connection must be based on the labelled base, collector, and emitter terminals, not only on the visual shape or orientation of the symbol.

3.7 Correct LED Polarity

The LED is a polarised component, so its direction in the circuit matters. The LED anode must be connected toward the positive supply side through RLED. The LED cathode must be connected toward the transistor collector.

This orientation allows conventional current to flow from the 5 V rail through RLED, through the LED from anode to cathode, and then through the transistor to ground when the transistor is on.

If the LED is reversed, the LED will block current under normal operating conditions. In that case, the transistor may be switching correctly, but the LED may still remain off. Therefore, LED polarity must be verified before interpreting the circuit as faulty.

The correct polarity rule is simple. The anode faces the supply through RLED. The cathode faces the collector of Q1.

3.8 Scheme Verification Checklist

Before starting the simulation, the scheme must be verified step by step. The 5 V source must be connected to the positive supply rail, and the negative terminal of the source must be connected to ground. The ground symbol must be present because Proteus requires a reference node for reliable circuit simulation.

R1 must be set to 10 k Ω and connected between the 5 V rail and the SENSE node. The LDR must be connected between the SENSE node and ground. This confirms that the divider orientation produces high SENSE voltage in darkness and low SENSE voltage in bright light.

The SENSE node must connect to RB, and RB must connect to the base of Q1. There must be no direct wire from SENSE to the base that bypasses RB. The emitter of Q1 must connect directly to ground. The collector of Q1 must connect to the LED cathode side.

RLED must be connected in series with the LED. The LED anode must connect toward RLED and the positive supply side. The LED cathode must connect toward the transistor collector. The voltmeter must be connected in parallel between SENSE and ground. The ammeter must be connected in series with the LED current path.

The verification process confirms that the scheme contains no missing current limiter, no floating ground reference, no reversed LED, no incorrect transistor connection, and no incorrect meter placement. Only after this checklist is satisfied should the circuit be simulated.

3.9 Scheme Summary

The circuit scheme is arranged as a light-controlled NPN transistor switch. The fixed resistor R1 and the LDR form the sensing stage. Their midpoint, called SENSE, produces a voltage that changes according to the light condition. The SENSE node is connected to the transistor base through RB, which limits the base current and protects the transistor.

The 2N2222 transistor is connected as a low side switch. Its emitter is connected to ground, its collector is connected to the LED cathode side, and its base receives the control signal through RB. The LED load is connected between the positive 5 V rail and the transistor collector through RLED and the LED. RLED limits the output current and protects both the LED and the transistor.

The scheme has three correct current paths. The divider current path produces the SENSE voltage. The base current path controls the transistor. The LED current path produces the visible output. In darkness, the LDR resistance increases, the SENSE voltage rises, base current flows, the transistor turns on, and the LED lights. In bright light, the LDR resistance decreases, the SENSE voltage falls, base current stops, the transistor turns off, and the LED remains off.

This chapter therefore defines the complete electrical arrangement required for a correct Proteus simulation. The scheme is valid because every signal path, control path, output path, current limiter, polarity condition, and measurement connection is explicitly defined.

Chapter 4

Simulation

4.1 Purpose of Simulation

The purpose of this chapter is to verify the electrical operation of the designed NPN transistor switching circuit before any physical construction is attempted. Simulation is necessary because it allows the circuit behavior to be tested under controlled light conditions, component values, voltage levels, and current paths. In this project, the simulation confirms whether the LDR voltage divider can produce a sufficient control voltage for the base of the 2N2222 transistor and whether the transistor can safely switch the LED current through the collector emitter path.

The simulation also serves as a validation stage between theoretical calculation and practical implementation. The calculations in the previous chapter predicted that the SENSE node voltage should become high in darkness and low in bright light. The simulation must therefore confirm that the circuit produces the same behavior in Proteus. If the simulated results match the calculated results within reasonable tolerance, the design can be considered electrically consistent and ready for physical construction.

This chapter has four main verification goals. First, it verifies that the voltage divider responds correctly to changes in LDR resistance. Second, it verifies that the base of the transistor receives sufficient voltage and current in the dark condition. Third, it verifies that the transistor enters the ON state in darkness and the OFF state in bright light. Fourth, it verifies that the LED current remains limited by the LED resistor and does not exceed the intended safe operating range.

Therefore, the simulation is not only a visual demonstration of the LED turning ON and OFF. It is a technical proof that the input stage, control stage, transistor stage, and output stage operate together as one correct electronic system.

4.2 Proteus Simulation Setup

The circuit was prepared in Proteus using the same component values selected during the calculation stage. The supply voltage was set to 5 volts. A 10 kilohm resistor and an LDR were connected as a voltage divider. The midpoint of the voltage divider was defined as the SENSE node. This SENSE node was connected to the base of the 2N2222 transistor through a 4.7 kilohm base resistor. The transistor was connected as a low side switch, with the emitter connected to ground and the collector connected to the LED load path. The LED was connected in series with a 470 ohm resistor to limit the load current.

The simulation uses two representative LDR resistance values. In the dark condition, the LDR resistance is taken as 100 kilohms. In the bright condition, the LDR resistance is taken as 1 kilohm. These two values allow the circuit to be tested at two extreme operating points. The dark condition represents high LDR resistance, high SENSE voltage, and transistor activation.

The bright condition represents low LDR resistance, low SENSE voltage, and transistor deactivation.

The component values used in the Proteus simulation are shown in Table 4.1.

Table 4.1

Proteus Simulation Component Values

Component	Simulation Value
DC supply V1	5 V
Fixed resistor R1	10 k Ω
LDR resistance in darkness	100 k Ω
LDR resistance in bright light	1 k Ω
Base resistor RB	4.7 k Ω
LED resistor RLED	470 Ω
Transistor Q1	2N2222
Output indicator	Red LED

The same values must be used throughout the simulation so that the simulated results can be directly compared with the theoretical calculations. Changing any of these values without recalculation would make the comparison invalid.

4.3 Simulation Instruments

The simulation requires measurement instruments so that the circuit behavior can be verified numerically, not only visually. A DC voltmeter is used to measure the voltage at the SENSE node with respect to ground. This measurement is essential because the SENSE voltage determines whether the transistor base receives enough voltage to conduct.

A DC ammeter is used to measure the current flowing through the LED and the transistor collector path. The ammeter must be connected in series with the load current path. It must not be connected in parallel because an ammeter has very low internal resistance, and a parallel connection would create an incorrect current path.

The LED itself is also used as a visual output indicator. When the transistor is ON, the LED should light. When the transistor is OFF, the LED should remain dark. However, the LED state

alone is not enough for academic verification. The voltmeter and ammeter readings are required to prove that the circuit works according to the calculated voltage and current values.

The required simulation instruments are listed in Table 4.2.

Table 4.2

Simulation Instruments and Their Functions

Instrument	Connection Point	Function
DC voltmeter	SENSE node to ground	Measures the voltage produced by the LDR divider
DC ammeter	In series with the LED current path	Measures the collector and LED current
Red LED	Output load path	Shows the visible switching state

The voltmeter confirms the input control condition. The ammeter confirms the output current condition. The LED confirms the observable output condition. Together, these three forms of evidence provide complete simulation verification.

4.4 Simulation Test 1

Dark Condition

4.4.1 Setup

For the first simulation test, the LDR resistance is set to approximately 100 kilohms. This represents the dark condition. In darkness, the LDR has high resistance, so it does not strongly pull the SENSE node toward ground. As a result, the SENSE voltage should rise close to the positive supply voltage.

The voltage divider consists of the 10 kilohm fixed resistor connected between the 5 volt supply and the SENSE node, and the 100 kilohm LDR connected between the SENSE node and ground. The SENSE voltage is calculated from the voltage divider relationship:

V_{SENSE} equals V_{CC} multiplied by R_{LDR} divided by the sum of R_1 and R_{LDR} .

Substituting the dark condition values gives:

VSENSE equals 5 multiplied by 100 kilohms divided by the sum of 10 kilohms and 100 kilohms.

VSENSE equals 500 divided by 110.

VSENSE is approximately 4.55 volts.

This voltage is greater than the approximate 0.7 volt base emitter threshold of a silicon NPN transistor. Therefore, base current should flow through the 4.7 kilohm base resistor.

The approximate base current is calculated as:

IB equals VSENSE minus VBE divided by RB.

Substituting the values gives:

IB equals 4.55 minus 0.7 divided by 4700.

IB equals 3.85 divided by 4700.

IB is approximately 0.000819 ampere.

IB is approximately 0.819 milliampere.

This base current is sufficient for the selected LED load current. Therefore, the transistor is expected to enter saturation and allow current to flow through the LED.

4.4.2 Expected Measurements

The expected simulation results for the dark condition are shown in Table 4.3.

Table 4.3

Expected Measurements in the Dark Condition

Quantity	Expected Value
LDR resistance	100 kΩ
SENSE voltage	approximately 4.55 V
Base current	approximately 0.819 mA
Collector current	approximately 5.96 mA

Quantity	Expected Value
Transistor state	ON
LED state	ON

The collector current is determined mainly by the 5 volt supply, the LED voltage, the collector emitter saturation voltage, and the 470 ohm LED resistor. The approximate collector current is calculated as:

I_C equals V_{CC} minus V_{LED} minus V_{CE} saturation divided by R_{LED} .

Substituting the selected values gives:

I_C equals 5 minus 2.0 minus 0.2 divided by 470.

I_C equals 2.8 divided by 470.

I_C is approximately 0.00596 ampere.

I_C is approximately 5.96 milliampere.

This value is suitable for a small red indicator LED and remains safely limited by the 470 ohm resistor.

4.4.3 Expected Explanation

In the dark condition, the LDR resistance is high. Because the LDR is connected between the SENSE node and ground, its high resistance reduces the amount of current pulled from SENSE to ground. This causes the SENSE voltage to rise to approximately 4.55 volts. The high SENSE voltage is applied to the transistor base through the 4.7 kiloohm base resistor.

Once the base emitter junction receives sufficient forward bias, base current flows from the SENSE node through R_B into the base, then through the emitter to ground. This base current activates the transistor. Since the transistor is connected as a low side switch, the collector emitter path provides a return path for the LED current.

The LED current flows from the 5 volt supply through the 470 ohm resistor, through the LED, into the transistor collector, out through the emitter, and finally to ground. The transistor is therefore in the ON state, and the LED lights. The expected simulation result is a visible red LED and a measured collector current close to 5.96 milliampere.

4.5 Simulation Test 2

Bright Condition

4.5.1 Setup

For the second simulation test, the LDR resistance is set to approximately 1 kilohm. This represents the bright light condition. In bright light, the LDR has low resistance, so it strongly pulls the SENSE node toward ground. As a result, the SENSE voltage should fall below the voltage needed to activate the transistor base emitter junction.

The voltage divider now consists of the same 10 kilohm fixed resistor connected between the 5 volt supply and the SENSE node, and a 1 kilohm LDR connected between the SENSE node and ground.

The SENSE voltage is calculated as:

V_{SENSE} equals V_{CC} multiplied by R_{LDR} divided by the sum of R_1 and R_{LDR} .

Substituting the bright condition values gives:

V_{SENSE} equals 5 multiplied by 1 kilohm divided by the sum of 10 kilohms and 1 kilohm.

V_{SENSE} equals 5 divided by 11.

V_{SENSE} is approximately 0.45 volt.

This voltage is lower than the approximate 0.7 volt needed to forward bias the base emitter junction. Therefore, the transistor should remain in the OFF state.

4.5.2 Expected Measurements

The expected simulation results for the bright condition are shown in Table 4.4.

Table 4.4

Expected Measurements in the Bright Condition

Quantity	Expected Value
LDR resistance	1 k Ω
SENSE voltage	approximately 0.45 V
Base current	approximately 0 mA

Quantity	Expected Value
Collector current	approximately 0 mA
Transistor state	OFF
LED state	OFF

Because the SENSE voltage is too low to activate the base emitter junction, the base current is expected to be approximately zero. Since the transistor does not turn ON, the collector emitter path remains open for practical switching purposes. As a result, the LED current is also approximately zero, and the LED remains OFF.

4.5.3 Expected Explanation

In bright light, the LDR resistance becomes low. Since the LDR is connected from the SENSE node to ground, this low resistance pulls the SENSE node toward ground. The resulting SENSE voltage is approximately 0.45 volt. This value is not sufficient to forward bias the base emitter junction of the 2N2222 transistor.

Because there is no significant base current, the transistor remains in the cutoff state. In cutoff, the collector emitter path does not provide a complete return path for the LED current. The load current therefore stops, and the LED remains OFF.

This behavior confirms that the LDR divider is correctly controlling the transistor. The same circuit that turns ON in darkness turns OFF in bright light because the SENSE voltage changes according to the LDR resistance.

4.6 Simulation Results Table

After completing both simulation tests, the measured results should be recorded in a single comparison table. This table allows the dark and bright conditions to be evaluated side by side. The table should include the LDR value, SENSE voltage, base current, collector current, transistor state, and LED state.

Table 4.5

Expected Simulation Results

Test Condition	LDR Value	SENSE Voltage	Base Current	Collector Current	Transistor State	LED State
Dark condition	100 k Ω	4.55 V	0.819 mA	5.96 mA	ON	ON
Bright condition	1 k Ω	0.45 V	approximately 0 mA	approximately 0 mA	OFF	OFF

The table demonstrates the complete switching behavior of the circuit. In darkness, the SENSE voltage rises, base current flows, collector current flows, and the LED turns ON. In bright light, the SENSE voltage falls, base current stops, collector current stops, and the LED turns OFF.

The results table is important because it proves that the circuit is not only visually working, but also numerically consistent with the theoretical design.

4.7 Comparison Between Calculation and Simulation

The calculated results and simulated results must be compared to confirm the validity of the circuit design. The most important comparison is the SENSE voltage because it directly determines the transistor switching state. In the dark condition, the calculated SENSE voltage is approximately 4.55 volts. The Proteus simulation should show a value very close to this result. If the simulated value is close to 4.55 volts, then the voltage divider has been built correctly.

In the bright condition, the calculated SENSE voltage is approximately 0.45 volt. The Proteus simulation should also show a value close to this result. If the simulated SENSE voltage is close to 0.45 volt, then the LDR divider is correctly pulling the base control voltage down in bright light.

The collector current must also be compared. In the dark condition, the calculated collector current is approximately 5.96 milliamperes. The simulation should show a collector current close to this value. Small differences may occur because the simulated LED forward voltage and transistor saturation voltage may not exactly equal the estimated values used in calculation. However, the current should remain in the same range and should clearly show that the LED current is limited by the 470 ohm resistor.

In the bright condition, the calculated collector current is approximately zero because the transistor is OFF. The simulation should also show approximately zero collector current. If collector current is still present in bright light, the likely causes are incorrect transistor wiring, incorrect LDR placement, incorrect base connection, or an unintended current path.

The LED state must also match the theoretical prediction. In darkness, the LED should be ON. In bright light, the LED should be OFF. This confirms that the circuit performs the intended light controlled switching function.

A complete comparison table may be written as shown in Table 4.6.

Table 4.6

Calculation and Simulation Comparison

Quantity	Calculated Dark Value	Expected Simulated Dark Value	Calculated Bright Value	Expected Simulated Bright Value
SENSE voltage	4.55 V	close to 4.55 V	0.45 V	close to 0.45 V
Base current	0.819 mA	close to 0.819 mA	approximately 0 mA	approximately 0 mA
Collector current	5.96 mA	close to 5.96 mA	approximately 0 mA	approximately 0 mA
Transistor state	ON	ON	OFF	OFF
LED state	ON	ON	OFF	OFF

If the simulated values agree with the calculated values, the design is validated. If the values do not agree, the circuit must be checked for incorrect resistor values, reversed LED polarity, missing ground connection, wrong transistor pin assignment, or incorrect LDR placement.

4.8 Required Simulation Screenshots

The academic report should include separate simulation screenshots as evidence. Each screenshot should prove one specific part of the circuit operation. The screenshots should not be combined into one unclear image because each image has a different verification purpose.

The first required screenshot is the full circuit scheme. This image must show the complete Proteus circuit, including the 5 volt supply, fixed resistor, LDR, SENSE node, base resistor, 2N2222 transistor, LED resistor, LED, voltmeter, ammeter, and ground. The purpose of this screenshot is to prove that the circuit was wired according to the designed schematic.

The second required screenshot is the dark condition simulation. This image must show the LDR set to the dark resistance value and the LED in the ON state. It should also show the voltmeter and ammeter readings if possible. The purpose of this screenshot is to prove that high LDR resistance produces high SENSE voltage and activates the transistor.

The third required screenshot is the bright condition simulation. This image must show the LDR set to the bright resistance value and the LED in the OFF state. The purpose of this screenshot is to prove that low LDR resistance pulls the SENSE voltage down and deactivates the transistor.

The fourth required screenshot is the SENSE voltage measurement. This image should focus on the DC voltmeter connected between the SENSE node and ground. The purpose of this screenshot is to prove the actual control voltage produced by the voltage divider.

The fifth required screenshot is the LED current measurement. This image should focus on the DC ammeter connected in series with the LED and collector current path. The purpose of this screenshot is to prove that the LED current is limited and matches the expected current range.

The sixth required screenshot is the troubleshooting case with the missing base resistor. This image is included only as a fault demonstration. It should show that removing the base resistor creates an unsafe or incorrect design condition. The purpose of this screenshot is to prove why the base resistor is necessary for protecting the transistor and controlling the base current.

The required screenshots are listed in Table 4.7.

Table 4.7

Required Simulation Screenshots

Screenshot Number	Screenshot Content	Purpose
1	Full circuit scheme	Shows complete Proteus wiring
2	Dark condition simulation	Shows LED ON behavior
3	Bright condition simulation	Shows LED OFF behavior
4	SENSE voltage measurement	Shows voltage divider output
5	LED current measurement	Shows limited load current
6	Missing base resistor case	Shows the importance of base current limiting

Each screenshot should have a figure number and a clear caption in the final report. The caption should explain what the image proves.

4.9 Simulation Accuracy Notes

Small differences may appear between calculated and simulated values. These differences are normal because the calculations use simplified component assumptions, while Proteus may use more detailed component models. For example, the calculation assumes that the red LED forward voltage is approximately 2.0 volts. In simulation, the actual LED model may use a slightly different forward voltage depending on the current. Similarly, the calculation assumes that the transistor collector emitter saturation voltage is approximately 0.2 volt. In simulation, this value may change slightly depending on the collector current and base current.

The base emitter voltage is also treated as approximately 0.7 volt in calculation. In simulation, this value may vary slightly depending on the transistor model and operating current. Therefore, the exact simulated base current may not be identical to the calculated base current. This does not mean the circuit is wrong. The design is considered correct if the simulated values are close to the calculated values and if the switching behavior is correct.

The most important simulation result is not perfect numerical equality. The most important result is correct electrical behavior. In the dark condition, the SENSE voltage must be high enough to turn the transistor ON, and the LED current must be safely limited. In the bright condition, the SENSE voltage must be low enough to keep the transistor OFF, and the LED current must be approximately zero.

Simulation accuracy also depends on correct meter placement. A voltmeter must be connected between two points to measure potential difference. In this project, the SENSE voltage is measured between the SENSE node and ground. An ammeter must be connected in series with the current path. In this project, the ammeter is placed in series with the LED and collector path. If the instruments are connected incorrectly, the simulation readings will not represent the real circuit behavior.

4.10 Simulation Summary

The Proteus simulation confirms the expected operation of the NPN transistor switching circuit. In the dark condition, the LDR resistance is high, causing the SENSE voltage to rise to approximately 4.55 volts. This voltage provides sufficient base current through the 4.7 kilohm resistor. The 2N2222 transistor turns ON, collector current flows through the LED path, and the LED lights. The expected LED current is approximately 5.96 milliampere, which is safely limited by the 470 ohm resistor.

In the bright condition, the LDR resistance is low, causing the SENSE voltage to fall to approximately 0.45 volt. This voltage is not sufficient to forward bias the base emitter junction. The transistor remains OFF, the collector current is approximately zero, and the LED remains OFF.

The simulation therefore verifies the full operating chain of the circuit. A change in light changes the LDR resistance. The LDR resistance changes the SENSE voltage. The SENSE voltage

controls the base current. The base current controls the transistor state. The transistor state controls the LED current. The LED current determines the visible output.

The agreement between calculation and simulation proves that the circuit is correctly designed, safely current limited, and suitable for implementation. The simulation also confirms the importance of the base resistor and the LED resistor. The base resistor protects the transistor input path, while the LED resistor protects the output current path. Without these two resistors, the circuit would no longer be electrically safe or academically correct.

Therefore, Chapter 4 establishes that the simulated circuit satisfies the intended project objective. The LDR voltage divider successfully controls the 2N2222 transistor, and the transistor successfully switches the LED output according to the light condition.

Chapter 5: Troubleshooting

5.1 Purpose of Troubleshooting

Troubleshooting is the systematic process of identifying, isolating, and correcting faults in the circuit. In this project, troubleshooting is necessary because the circuit depends on correct interaction between the LDR voltage divider, the base resistor, the 2N2222 transistor, the LED load, the current limiting resistor, the power supply, and the ground reference. A fault in any one of these parts can prevent the circuit from operating correctly.

The purpose of this chapter is to explain the most likely errors that may occur during the Proteus simulation and to provide precise correction methods for each error. The troubleshooting process focuses on three main fault types: wiring errors, component value errors, and simulation setup errors. Wiring errors include incorrect transistor pin connections, reversed LED polarity, missing ground, or incorrect meter placement. Component value errors include using an incorrect base resistor, incorrect LED resistor, incorrect LDR resistance value, or incorrect supply voltage. Simulation setup errors include missing ground symbols, floating circuit nodes, invalid component models, and incorrect instrument connections.

The correct operation of the circuit is based on clear expected values. In darkness, the LDR resistance is assumed to be approximately 100 k Ω , and the SENSE voltage should be approximately 4.55 V. In bright light, the LDR resistance is assumed to be approximately 1 k Ω , and the SENSE voltage should be approximately 0.45 V. The LED current when the transistor is ON should be approximately 5.96 mA. The base current in darkness should be approximately 0.819 mA when the base resistor is 4.7 k Ω . These values provide the standard reference for all troubleshooting decisions.

5.2 Problem 1: Transistor Gets Hot

A transistor that becomes hot indicates that excessive current may be flowing through one of its terminals. In this project, the 2N2222 transistor should not become hot because the circuit is designed for a small LED load current of approximately 5.96 mA. If the transistor heats up, the most likely reason is that the current has not been properly limited.

5.2.1 Possible Causes

The first possible cause is that the base resistor is missing. The base resistor is essential because the base emitter junction behaves similarly to a diode and must not be connected directly to the SENSE node without current limitation. If the base resistor is removed, excessive base current may flow when the SENSE voltage is high.

The second possible cause is that the LED resistor is missing. The LED resistor limits the collector current. If this resistor is removed, the LED and transistor may draw excessive current from the 5 V source, which can cause overheating or unrealistic simulation behavior.

The third possible cause is incorrect transistor pin connection. If the base, collector, and emitter are connected wrongly, the transistor may not operate as a normal low side switch. Incorrect pin wiring can force current through unintended internal paths.

The fourth possible cause is a collector emitter short circuit. If the collector and emitter are accidentally connected together, the load current may bypass the normal transistor switching behavior.

The fifth possible cause is excessive load current. This can occur if the LED resistor value is too small or if another load is connected without proper current limitation.

5.2.2 Correction

The correction process must begin by verifying the two current limiting resistors. The base resistor must be installed with a value of 4.7 k Ω . This resistor must be connected between the SENSE node and the transistor base. The LED resistor must be installed with a value of 470 Ω . This resistor must be placed in series with the LED.

The transistor pins must then be checked carefully. The base must be connected only to the base resistor. The collector must be connected to the LED cathode. The emitter must be connected directly to ground. If the transistor symbol or model has a different pin arrangement, the Proteus pin information must be checked before running the simulation again.

The expected collector current should also be verified. Using the selected values, the approximate LED current is calculated as follows:

IC equals 5 V minus 2.0 V minus 0.2 V, all divided by 470 Ω .

IC equals 2.8 V divided by 470 Ω .

IC is approximately 0.00596 A.

IC is approximately 5.96 mA.

This current is small and safe for the transistor. Therefore, if the transistor still becomes hot after RB equals 4.7 k Ω and RLED equals 470 Ω are installed correctly, the fault is most likely a wiring error or an incorrect transistor model.

5.3 Problem 2: LED Always OFF

If the LED is always OFF, the transistor is probably not receiving enough base drive, or the load path is incomplete. The LED should turn ON in darkness because the LDR resistance becomes high, causing the SENSE voltage to rise. If the LED does not turn ON in darkness, the first diagnostic target must be the SENSE node.

5.3.1 Possible Causes

The first possible cause is that the SENSE voltage is too low. If the SENSE voltage does not rise above the approximate base emitter turn on voltage of 0.7 V, the transistor will not conduct.

The second possible cause is that the LDR and R1 are swapped. In the intended circuit, R1 is connected between the 5 V rail and the SENSE node, while the LDR is connected between the SENSE node and ground. This orientation makes the SENSE voltage high in darkness and low in bright light. If the two components are exchanged, the behavior can be reversed.

The third possible cause is that the base resistor is not connected to the SENSE node. If the SENSE node is correct but the base resistor is connected elsewhere, the transistor base will not receive the intended control signal.

The fourth possible cause is that the emitter is not connected to ground. The emitter must have a direct ground reference so that the base emitter junction can become forward biased.

The fifth possible cause is reversed LED polarity. The LED anode must face the positive supply through the 470 Ω resistor, and the LED cathode must connect toward the transistor collector.

The sixth possible cause is incorrect transistor pin wiring. If the collector, base, and emitter are not connected to the correct nodes, the transistor may remain OFF even when the SENSE voltage is correct.

5.3.2 Correction

The correction must begin by measuring the SENSE voltage in darkness. The LDR should be set to approximately 100 k Ω . With R1 equal to 10 k Ω and the supply voltage equal to 5 V, the SENSE voltage should be approximately 4.55 V.

The calculation is:

V_{SENSE} equals 5 V multiplied by 100 k Ω , divided by 10 k Ω plus 100 k Ω .

V_{SENSE} equals 500 k Ω volts divided by 110 k Ω .

V_{SENSE} is approximately 4.55 V.

If this voltage is not present, the divider orientation must be checked. R1 must connect from the 5 V rail to the SENSE node. The LDR must connect from the SENSE node to ground.

If the SENSE voltage is correct, the next check is the base resistor. RB must connect from the SENSE node to the transistor base. The transistor emitter must be connected to ground. The LED anode must connect toward the 5 V rail through RLED, and the LED cathode must connect to the transistor collector.

If all these connections are correct, the base current should be sufficient. With V_{SENSE} equal to approximately 4.55 V, V_{BE} equal to approximately 0.7 V, and RB equal to 4.7 k Ω , the base current is:

I_B equals 4.55 V minus 0.7 V, divided by 4700 Ω .

I_B equals 3.85 V divided by 4700 Ω .

I_B is approximately 0.000819 A.

I_B is approximately 0.819 mA.

This is greater than the required base current of approximately 0.596 mA for a 5.96 mA collector current under the forced switching condition. Therefore, if the LED remains OFF despite this base current, the fault is most likely in the LED polarity, collector connection, emitter connection, or transistor pin mapping.

5.4 Problem 3: LED Always ON

If the LED is always ON, the transistor is remaining in conduction even when it should be OFF. In the intended design, bright light should reduce the LDR resistance, pull the SENSE node down, remove base drive, and turn the transistor OFF. A permanently ON LED means that the base is still being driven, the transistor path is bypassed, or the transistor pins are incorrectly connected.

5.4.1 Possible Causes

The first possible cause is that the SENSE voltage is always high. This may occur if the LDR is not connected correctly or if its resistance is not changing during simulation.

The second possible cause is that the LDR value is not changing. If the LDR remains at a high resistance value, the SENSE node will remain high, and the transistor will stay ON.

The third possible cause is that the base is accidentally connected directly to 5 V. This would force continuous base current regardless of the LDR state.

The fourth possible cause is that the collector and emitter are shorted. If this occurs, the LED current may flow even when the transistor is not being controlled properly.

The fifth possible cause is wrong transistor pin mapping. If the collector and emitter are interchanged or if the base is connected to the wrong terminal, the circuit may not respond to the LDR signal correctly.

5.4.2 Correction

The first correction is to set the LDR to the bright condition. The LDR resistance should be approximately 1 k Ω . With R1 equal to 10 k Ω and the LDR connected to ground, the SENSE voltage should fall to approximately 0.45 V.

The calculation is:

V_{SENSE} equals 5 V multiplied by 1 k Ω , divided by 10 k Ω plus 1 k Ω .

V_{SENSE} equals 5 k Ω volts divided by 11 k Ω .

V_{SENSE} is approximately 0.45 V.

This voltage is lower than the approximate 0.7 V needed to forward bias the base emitter junction. Therefore, the transistor should turn OFF.

If the SENSE voltage does not fall to approximately 0.45 V, the divider wiring or LDR setting is incorrect. The LDR must be confirmed as the lower resistor connected between SENSE and ground.

If the SENSE voltage is correct but the LED remains ON, the base wiring must be checked. The base must not be connected directly to 5 V. It must receive its signal only through the 4.7 k Ω resistor from the SENSE node.

The transistor pinout must also be checked. The emitter must connect to ground, and the collector must connect to the LED cathode. Finally, the load path must be checked to confirm that LED current is not bypassing the transistor through an unintended wire.

5.5 Problem 4: LED Is Too Dim

A dim LED indicates that some current is flowing, but the current is lower than expected. In this circuit, the LED should receive approximately 5.96 mA when the transistor is ON. If the LED is weak, the collector current may be too low, the base current may be insufficient, or the transistor may not be entering saturation.

5.5.1 Possible Causes

The first possible cause is that the LED resistor is too large. A larger resistor reduces the LED current and causes weaker brightness.

The second possible cause is that the base resistor is too large. If R_B is too large, the base current may be insufficient to drive the transistor into saturation.

The third possible cause is that the SENSE voltage is not high enough in darkness. If the LDR resistance is not sufficiently high, or if the divider is connected incorrectly, the base may receive only partial drive.

The fourth possible cause is that the transistor is not fully ON. This occurs when the available base current is lower than the amount required for reliable switching.

The fifth possible cause is that the wrong LED model is used. Some LED models may have a higher forward voltage or may appear dimmer under the same current conditions.

5.5.2 Correction

The first correction is to verify that R_{LED} equals 470 Ω . This resistor produces an expected LED current of approximately 5.96 mA when the transistor is saturated.

The second correction is to verify that R_B equals 4.7 k Ω . This value allows the base current to reach approximately 0.819 mA in darkness.

The dark state SENSE voltage must then be measured. It should be approximately 4.55 V. If it is much lower, the LDR divider must be checked.

The base current must also be compared with the switching requirement. The expected collector current is approximately 5.96 mA. For reliable switching, the base current target is approximately one tenth of the collector current.

I_B required equals 5.96 mA divided by 10.

I_B required is approximately 0.596 mA.

The actual base current in darkness is approximately 0.819 mA. Since 0.819 mA is greater than 0.596 mA, the transistor should be fully ON. If the LED remains dim after these values are verified, the fault is likely related to LED model selection, incorrect LED polarity, poor collector connection, or wrong transistor pin mapping.

5.6 Problem 5: Proteus Simulation Does Not Run

If the Proteus simulation does not run, the problem is usually not the theoretical circuit design. It is usually caused by a missing reference node, a floating connection, an invalid component, or incorrect instrument placement. A circuit simulator requires a complete electrical reference and valid current paths before it can solve the circuit.

5.6.1 Possible Causes

The first possible cause is missing ground. Every Proteus circuit must include a ground symbol because the simulator needs a zero volt reference point.

The second possible cause is a floating node. A floating node is a point in the circuit that is not properly connected to a defined voltage reference or current path.

The third possible cause is incorrect ammeter connection. An ammeter must be placed in series with the current path. If it is connected in parallel, it may create an incorrect low resistance path.

The fourth possible cause is a wrong power source setting. The DC source must be set to 5 V for this project.

The fifth possible cause is an invalid component model. If the selected transistor, LDR, LED, or meter model is not suitable for simulation, Proteus may fail to run or produce unreliable results.

5.6.2 Correction

The first correction is to add a ground symbol. The negative terminal of the 5 V source must connect to ground. The transistor emitter must also connect to ground.

All circuit nodes must then be checked for proper connection. The SENSE node must connect to R1, the LDR, and RB. The base resistor must connect to the transistor base. The LED path must connect from the 5 V rail through RLED and the LED to the transistor collector.

The ammeter must be placed in series with the LED current path. A correct placement is between the LED cathode and the transistor collector. The voltmeter must be connected across the measured voltage. For SENSE voltage, the voltmeter positive terminal must connect to SENSE, and the voltmeter negative terminal must connect to ground.

The DC source must be set to 5 V. If a component model is unclear or causes errors, it should be replaced with a standard Proteus component such as a standard resistor, standard LED, standard 2N2222 transistor, standard DC source, and standard ground.

5.7 Problem 6: Measurement Values Are Wrong

Incorrect measurement values can occur even when the circuit itself is mostly correct. In this project, the main measured values are SENSE voltage and LED current. A wrong reading usually means that the meter is connected incorrectly, the LDR setting is wrong, resistor values are wrong, or the ground reference is incorrect.

5.7.1 Possible Causes

The first possible cause is incorrect voltmeter connection. A voltmeter must be connected in parallel with the point being measured. To measure SENSE voltage, the voltmeter must be connected between SENSE and ground.

The second possible cause is incorrect ammeter connection. An ammeter must be placed in series with the current being measured. If it is placed in parallel, the measured value will be wrong and the circuit operation may be disturbed.

The third possible cause is wrong LDR resistance setting. If the LDR is not set to approximately 100 k Ω for darkness or approximately 1 k Ω for bright light, the measured SENSE voltage will not match the expected values.

The fourth possible cause is wrong resistor values. If R1, RB, or RLED has the wrong value, both the calculated and simulated values will change.

The fifth possible cause is a wrong ground reference. If the voltmeter or circuit is not referenced to the same ground node, the measured values may be misleading.

5.7.2 Correction

The voltmeter must be connected with its positive terminal at the SENSE node and its negative terminal at ground. This gives the correct SENSE voltage relative to the circuit reference.

The ammeter must be placed in series with the LED current path. A suitable placement is between the LED cathode and the transistor collector. This allows the ammeter to measure the same current that flows through the LED and the transistor collector.

All resistor values must be checked. R1 must be 10 k Ω . RB must be 4.7 k Ω . RLED must be 470 Ω . The LDR must be checked in both conditions. In darkness it should be approximately 100 k Ω . In bright light it should be approximately 1 k Ω .

After these checks, the expected measurements should be as follows. In darkness, the SENSE voltage should be approximately 4.55 V, the base current should be approximately 0.819 mA, and the collector current should be approximately 5.96 mA. In bright light, the SENSE voltage should be approximately 0.45 V, the base current should be approximately zero, and the collector current should be approximately zero.

5.8 Troubleshooting Table

The following table summarizes the main faults, their most likely causes, the first measurement or inspection step, and the required correction.

Problem	Most Likely Cause	First Check	Correction
Transistor hot	Missing RB or RLED	Base resistor and LED resistor	Add RB equal to 4.7 k Ω and RLED equal to 470 Ω
LED always OFF	No base drive	SENSE voltage in darkness	Fix the divider wiring and base resistor connection
LED always ON	Base always high	SENSE voltage in bright light	Fix the LDR divider and remove any direct base connection to 5 V
LED dim	Low collector current or weak base drive	RLED, RB, and SENSE voltage	Verify calculations and correct resistor values
Simulation fails	Missing ground or floating node	Ground symbol and node connections	Add ground and complete all circuit paths
Wrong readings	Wrong meter placement	Voltmeter and ammeter connections	Reconnect voltmeter in parallel and ammeter in series

This table should be used after the circuit has been checked against the final scheme. It is not a replacement for understanding the circuit. It is a condensed diagnostic guide that helps identify the most likely failure quickly.

5.9 Troubleshooting Summary

Most faults in this project come from a small number of structural causes. The first major cause is missing ground. Without a ground reference, Proteus cannot correctly solve the circuit, and the transistor emitter has no defined reference point.

The second major cause is a missing base resistor. Without RB, the base current is not safely limited. This can produce excessive current and unrealistic transistor behavior.

The third major cause is a missing LED resistor. Without RLED, the LED current and collector current are not properly limited, which can damage components in a real circuit and create invalid simulation results.

The fourth major cause is wrong transistor pinout. The 2N2222 must be connected so that the base receives the control signal, the collector connects to the LED load, and the emitter connects to ground. If these pins are interchanged, the circuit may fail, remain always ON, remain always OFF, or produce incorrect measurements.

The fifth major cause is wrong LDR divider orientation. In the intended design, R1 connects from the 5 V rail to the SENSE node, and the LDR connects from SENSE to ground. This creates a high SENSE voltage in darkness and a low SENSE voltage in bright light. If this orientation is changed, the circuit behavior may reverse.

The final diagnostic conclusion is that the circuit can be considered correct only when five conditions are simultaneously satisfied. First, the circuit has a common ground. Second, the SENSE node changes from approximately 4.55 V in darkness to approximately 0.45 V in bright light. Third, the base current is limited by a 4.7 k Ω resistor. Fourth, the LED current is limited by a 470 Ω resistor. Fifth, the 2N2222 transistor is connected with its base, collector, and emitter in the correct positions. When these five conditions are satisfied, the circuit should operate as a stable light controlled transistor switch.

Chapter 6 — Questions

6.1 Conceptual Questions

This section evaluates the theoretical understanding of the circuit. The purpose of these questions is to confirm that the working principle of the light-controlled transistor circuit is understood before discussing numerical values, wiring details, simulation behavior, and fault correction. Each answer explains the function of a major component and its role in the complete electronic system.

6.1.1 What is the purpose of the transistor in this circuit?

The purpose of the transistor in this circuit is to allow a small control signal from the LDR voltage divider to control a larger current through the LED load. The LDR voltage divider by itself produces a voltage signal, but it is not intended to supply a larger load current directly. The 2N2222 NPN transistor solves this limitation by acting as an electronic switching device. When sufficient base current is supplied, the transistor allows current to flow from the collector to the emitter, completing the LED current path. Therefore, the transistor separates the sensing function from the load-driving function and makes the circuit more reliable, safer, and more effective.

6.1.2 Why is the 2N2222 used?

The 2N2222 is used because it is a common general-purpose NPN bipolar junction transistor suitable for low-current switching and amplification tasks. In this project, it is used as a low-side switch that controls the LED current. The transistor is appropriate because the LED current is small, the supply voltage is low, and the required base current can be safely controlled using a resistor. Its use also makes the project educationally valuable because it clearly demonstrates the relationship between base current and collector current in a practical switching circuit.

6.1.3 What is the function of the LDR?

The function of the LDR is to sense the level of light by changing its resistance. When the surrounding light is bright, the resistance of the LDR becomes low. When the surrounding light is dark, the resistance of the LDR becomes high. In this circuit, the LDR is not used as a power component. It is used as a sensing component. Its resistance change is converted into a voltage change by the voltage divider, and that voltage is then used to control the base of the transistor.

6.1.4 What is the function of the voltage divider?

The function of the voltage divider is to convert the changing resistance of the LDR into a changing voltage at the SENSE node. The voltage divider is made from a fixed resistor and the LDR. In this project, the fixed resistor is connected between the positive supply and the SENSE node, while the LDR is connected between the SENSE node and ground. This arrangement causes the SENSE voltage to become high in darkness and low in bright light. The SENSE voltage is then used as the input control voltage for the transistor base through the base resistor.

The voltage divider relation is:

$$V_SENSE = V_CC \times R_LDR \div (R1 + R_LDR)$$

This formula shows that the SENSE voltage depends directly on the LDR resistance. As the LDR resistance increases, the SENSE voltage increases. As the LDR resistance decreases, the SENSE voltage decreases.

6.1.5 What is the function of the base resistor?

The function of the base resistor is to limit the current entering the base of the transistor. The base and emitter of an NPN transistor behave similarly to a diode junction when forward biased. If the SENSE node were connected directly to the base without a resistor, excessive current could flow into the base. This could overheat the transistor, damage the component, or produce incorrect simulation behavior. The base resistor therefore provides electrical protection and ensures controlled transistor operation.

In this circuit, the base resistor has a value of 4.7 kΩ. It allows enough base current to turn the transistor on in darkness while preventing unsafe current flow into the base.

6.1.6 What is the function of the LED resistor?

The function of the LED resistor is to limit the current flowing through the LED and the collector path of the transistor. Without this resistor, the LED could draw excessive current from the supply, which could damage the LED and place unnecessary stress on the transistor. The LED resistor controls the load current and makes the output stage safe.

In this circuit, the LED resistor has a value of 470 Ω. With a 5 V supply, a red LED voltage of approximately 2.0 V, and a transistor saturation voltage of approximately 0.2 V, this resistor limits the LED current to approximately 5.96 mA.

6.2 Calculation Questions

This section verifies the mathematical correctness of the design. The calculations prove that the selected resistor values produce the required transistor behavior in both darkness and bright light. The purpose of this section is to show that the circuit is not only visually correct in Proteus, but also electrically justified by numerical analysis.

6.2.1 What is the SENSE voltage in darkness?

In darkness, the LDR resistance is assumed to be approximately 100 kΩ. The fixed resistor R1 has a value of 10 kΩ, and the supply voltage is 5 V. Since the LDR is connected between the SENSE node and ground, the voltage divider formula is:

$$V_{\text{SENSE}} = V_{\text{CC}} \times R_{\text{LDR}} \div (R1 + R_{\text{LDR}})$$

Substituting the dark condition values gives:

$$V_{\text{SENSE}} = 5 \times 100 \text{ k}\Omega \div (10 \text{ k}\Omega + 100 \text{ k}\Omega)$$

$$V_{\text{SENSE}} = 500 \div 110$$

$$V_{\text{SENSE}} \approx 4.55 \text{ V}$$

Therefore, the SENSE voltage in darkness is approximately 4.55 V. This voltage is high enough to forward bias the base and emitter junction of the transistor through the base resistor.

6.2.2 What is the SENSE voltage in bright light?

In bright light, the LDR resistance is assumed to be approximately 1 k Ω . The fixed resistor remains 10 k Ω , and the supply voltage remains 5 V. Using the same voltage divider formula:

$$V_{\text{SENSE}} = V_{\text{CC}} \times R_{\text{LDR}} \div (R_1 + R_{\text{LDR}})$$

Substituting the bright condition values gives:

$$V_{\text{SENSE}} = 5 \times 1 \text{ k}\Omega \div (10 \text{ k}\Omega + 1 \text{ k}\Omega)$$

$$V_{\text{SENSE}} = 5 \div 11$$

$$V_{\text{SENSE}} \approx 0.45 \text{ V}$$

Therefore, the SENSE voltage in bright light is approximately 0.45 V. This value is below the approximate 0.7 V needed to forward bias the base and emitter junction. As a result, the transistor remains off.

6.2.3 What is the approximate LED current?

The approximate LED current is calculated from the supply voltage, the LED forward voltage, the transistor saturation voltage, and the LED series resistor. The relevant values are:

$$V_{\text{CC}} = 5 \text{ V}$$

$$V_{\text{LED}} \approx 2.0 \text{ V}$$

$$V_{\text{CE(sat)}} \approx 0.2 \text{ V}$$

$$R_{\text{LED}} = 470 \text{ }\Omega$$

The current is calculated as:

$$I_{\text{C}} = (V_{\text{CC}} - V_{\text{LED}} - V_{\text{CE(sat)}}) \div R_{\text{LED}}$$

Substituting the values gives:

$$I_{\text{C}} = (5 - 2.0 - 0.2) \div 470$$

$$I_{\text{C}} = 2.8 \div 470$$

$$I_{\text{C}} \approx 0.00596 \text{ A}$$

$$I_{\text{C}} \approx 5.96 \text{ mA}$$

Therefore, the LED current is approximately 5.96 mA when the transistor is on. This is a suitable current for an indicator LED and is safely limited by the 470 Ω resistor.

6.2.4 What is the required base current for switching?

For reliable transistor switching, the transistor should be driven into saturation. A practical forced switching condition is to use a base current approximately equal to one tenth of the collector current. This gives the condition:

$$I_{B(\text{required})} = I_C \div 10$$

Since the calculated collector current is approximately 5.96 mA:

$$I_{B(\text{required})} = 5.96 \text{ mA} \div 10$$

$$I_{B(\text{required})} \approx 0.596 \text{ mA}$$

Therefore, the required base current for reliable switching is approximately 0.596 mA. This value is used to check whether the selected base resistor provides enough current to turn the transistor fully on.

6.2.5 What is the actual base current in darkness?

In darkness, the SENSE voltage is approximately 4.55 V. The approximate base and emitter junction voltage is 0.7 V, and the base resistor is 4.7 kΩ. The base current is calculated as:

$$I_B = (V_{\text{SENSE}} - V_{BE}) \div R_B$$

Substituting the values gives:

$$I_B = (4.55 - 0.7) \div 4700$$

$$I_B = 3.85 \div 4700$$

$$I_B \approx 0.000819 \text{ A}$$

$$I_B \approx 0.819 \text{ mA}$$

Therefore, the actual base current in darkness is approximately 0.819 mA. This is greater than the required base current of approximately 0.596 mA. Therefore, the transistor should switch on correctly in darkness.

6.3 Scheme Questions

This section verifies the physical wiring of the circuit. A correct calculation is not enough if the circuit is wired incorrectly. Therefore, these questions confirm the exact placement of the

transistor terminals, LED, resistors, LDR, supply, and ground. The main purpose is to ensure that the Proteus scheme matches the intended electronic design.

6.3.1 Where is the emitter connected?

The emitter of the 2N2222 transistor is connected directly to ground. This connection is essential because the transistor is used as a low-side switch. The emitter provides the return path for both the base current and the collector current. If the emitter is not connected to ground, the transistor does not have a stable reference point and the circuit may not operate correctly.

6.3.2 Where is the collector connected?

The collector is connected to the cathode of the LED. This places the transistor below the LED load in the current path. When the transistor turns on, current flows from the positive supply through the LED resistor, then through the LED, then into the collector, then out through the emitter to ground. This is the correct arrangement for an NPN transistor used as a low-side switch.

6.3.3 Where is the base connected?

The base is connected to the SENSE node through the 4.7 k Ω base resistor. The SENSE node provides the light-dependent control voltage produced by the voltage divider. The base resistor is required between SENSE and the transistor base because it limits the base current. The base must not be connected directly to the SENSE node, because that would remove the current-limiting protection.

6.3.4 Where is the LED anode connected?

The LED anode is connected to the positive 5 V supply through the 470 Ω resistor. This means that the LED receives supply voltage through a current-limiting resistor. The resistor must be placed in series with the LED so that the LED current is controlled. The LED anode must face the positive side of the circuit, while the cathode must face the transistor collector.

6.3.5 Where is the LDR connected?

The LDR is connected between the SENSE node and ground. The fixed resistor R1 is connected between the positive supply and the SENSE node. This arrangement creates a voltage divider in which the SENSE voltage rises when the LDR resistance increases. Therefore, in darkness, the LDR resistance is high and the SENSE voltage becomes high enough to turn on the transistor. In bright light, the LDR resistance is low and the SENSE voltage falls, turning the transistor off.

6.4 Simulation Questions

This section verifies the expected behavior of the circuit in Proteus. The simulation must confirm that the mathematical predictions and the wiring structure produce the correct practical result. The simulation questions focus on two main conditions: darkness and bright light.

6.4.1 What should happen in darkness?

In darkness, the resistance of the LDR increases. Since the LDR is connected between the SENSE node and ground, this higher resistance causes the SENSE voltage to rise. The calculated SENSE voltage in darkness is approximately 4.55 V. This voltage supplies base current through the 4.7 k Ω base resistor. The base current turns on the 2N2222 transistor, allowing collector current to flow through the LED path. Therefore, the LED should turn on in darkness.

The expected sequence is:

Darkness causes high LDR resistance

High LDR resistance causes high SENSE voltage

High SENSE voltage causes base current

Base current turns on the transistor

The transistor completes the LED current path

The LED turns on

6.4.2 What should happen in bright light?

In bright light, the resistance of the LDR decreases. Because the LDR is connected from the SENSE node to ground, this low resistance pulls the SENSE voltage down. The calculated SENSE voltage in bright light is approximately 0.45 V. This voltage is not high enough to forward bias the base and emitter junction of the transistor. Therefore, base current is approximately zero, the transistor remains off, collector current does not flow, and the LED stays off.

The expected sequence is:

Bright light causes low LDR resistance

Low LDR resistance causes low SENSE voltage

Low SENSE voltage prevents base current

The transistor remains off

The LED current path remains open

The LED turns off

6.4.3 What should the voltmeter show in darkness?

In darkness, the voltmeter should show approximately 4.55 V when connected between the SENSE node and ground. This value confirms that the LDR voltage divider is working correctly in the dark condition. If the measured value is much lower than 4.55 V, then the LDR value, resistor placement, supply voltage, or ground connection should be checked.

6.4.4 What should the voltmeter show in bright light?

In bright light, the voltmeter should show approximately 0.45 V when connected between the SENSE node and ground. This value confirms that the LDR resistance has decreased and that the SENSE node is being pulled close to ground. If the measured value remains high in bright light, then the LDR setting, LDR connection, or voltage divider orientation should be checked.

6.4.5 What should the ammeter show when the LED is on?

When the LED is on, the ammeter placed in series with the LED and collector path should show approximately 5.96 mA. This confirms that the LED current is being properly limited by the 470 Ω resistor. If the measured current is much higher, then the LED resistor may be missing or incorrectly valued. If the measured current is near zero while the circuit is supposed to be on, then the transistor wiring, LED polarity, base resistor connection, or ground connection should be checked.

6.5 Troubleshooting Questions

This section checks whether faults can be identified logically. Troubleshooting is not based on guessing. It is based on verifying the required electrical paths one by one. A working circuit must have a correct supply path, a correct ground reference, a correct voltage divider, a protected base path, and a protected LED current path.

6.5.1 What should be checked first if the transistor becomes hot?

If the transistor becomes hot, the first components to check are the base resistor and the LED resistor. The base resistor protects the base and emitter junction by limiting base current. The LED resistor protects the collector path by limiting load current. If either resistor is missing, incorrectly connected, or too small in value, excessive current may flow. This can cause heating, incorrect simulation behavior, or damage in a real circuit.

The first checks should be:

$R_B = 4.7 \text{ k}\Omega$

$R_{LED} = 470 \text{ }\Omega$

Emitter connected to ground

Collector connected to the LED cathode

Base connected only through R_B

6.5.2 What should be checked if the LED is always off?

If the LED is always off, the first measurement should be the SENSE voltage in darkness. In the dark condition, SENSE should be approximately 4.55 V. If SENSE does not rise to this value, then the voltage divider is not producing the required control voltage. The LDR and fixed resistor orientation should be checked. If SENSE is correct but the LED is still off, then the base resistor connection, transistor pinout, emitter ground connection, LED polarity, and collector connection should be checked.

The diagnostic order should be:

Measure SENSE in darkness

Confirm that SENSE is approximately 4.55 V

Confirm that SENSE reaches R_B

Confirm that R_B reaches the base

Confirm that the emitter reaches ground

Confirm that the LED cathode reaches the collector

Confirm that the LED anode reaches the positive supply through R_{LED}

6.5.3 What should be checked if the LED is always on?

If the LED is always on, the first measurement should be the SENSE voltage in bright light. In the bright condition, SENSE should fall to approximately 0.45 V. If SENSE remains high in bright light, then the LDR is not pulling the SENSE node down correctly. This may happen if the LDR value is not changing, the LDR is not connected to ground, the fixed resistor and LDR are incorrectly arranged, or the base is accidentally connected to the positive supply.

If SENSE is low but the LED remains on, then the fault is likely in the transistor or output stage. The collector and emitter may be incorrectly connected, the transistor model may have an unexpected pin arrangement, or the LED path may be bypassing the transistor.

6.5.4 What should be checked if Proteus gives incorrect behavior?

If Proteus gives incorrect behavior, the first checks should be ground, transistor pinout, resistor values, and meter connections. A missing ground is one of the most common simulation errors because the circuit needs a zero-volt reference point. The transistor pinout must also be verified because base, collector, and emitter connections are not interchangeable. The resistor values must match the calculation values. The voltmeter must be placed in parallel between SENSE and ground, while the ammeter must be placed in series with the LED current path.

The required checks are:

Ground must be present

Supply must be 5 V

R1 must be 10 k Ω

R_B must be 4.7 k Ω

R_{LED} must be 470 Ω

The transistor pins must match base, collector, and emitter

The voltmeter must measure SENSE relative to ground

The ammeter must be in series with the LED current path

6.5.5 What mistake causes excessive base current?

Excessive base current is caused by connecting the SENSE node directly to the transistor base without the base resistor. This removes the current-limiting component from the base path. Since the base and emitter junction behaves like a forward-biased diode when the transistor is on, current can become too large if it is not limited by a resistor. The correct connection is:

SENSE node to R_B

R_B to transistor base

The direct connection:

SENSE node to transistor base

is incorrect because it does not safely control the base current.

6.6 Final Understanding Questions

This section confirms complete understanding of the project. These questions connect the concept, calculation, scheme, simulation, and troubleshooting into one final explanation. The purpose is to prove that the student understands not only what the circuit does, but also why it works and why each component is necessary.

6.6.1 Why is the circuit called an electronic gate?

The circuit is called an electronic gate because the transistor controls whether current is allowed to pass through the LED path. The base signal acts as the control input. When the base receives sufficient current, the transistor turns on and permits collector current to flow. When the base does not receive sufficient current, the transistor remains off and blocks the LED current path. Therefore, the transistor behaves like an electrically controlled gate for current.

6.6.2 Why is the LDR not connected directly to the LED?

The LDR is not connected directly to the LED because the LDR voltage divider is a sensing stage, not a proper load-driving stage. Its purpose is to produce a voltage that changes with light. If the LED were driven directly from the LDR divider, the LED current would depend strongly on the LDR and divider values, and the circuit would not have a clean switching action. By using the transistor, the LDR stage only controls the base, while the LED current is supplied through a separate load path. This improves reliability and makes the circuit easier to calculate, simulate, and troubleshoot.

6.6.3 Why must the circuit share one common ground?

The circuit must share one common ground because the voltage divider, transistor base, transistor emitter, supply, and measurement instruments all need the same voltage reference. The SENSE voltage is meaningful only when measured relative to ground. The base and emitter voltage is also meaningful only when both points share the same reference system. If the voltage divider ground and the transistor emitter ground are not common, the transistor may not receive the intended base and emitter voltage, and the circuit may fail even if the schematic appears partly connected.

6.6.4 What proves that the circuit is correct?

The circuit is proved correct when the calculation, scheme, and simulation all agree. The calculations show that the SENSE voltage is approximately 4.55 V in darkness and approximately 0.45 V in bright light. They also show that the LED current is approximately 5.96 mA and that the dark-state base current is approximately 0.819 mA, which is greater than the required base current of approximately 0.596 mA. The scheme shows that every component is

connected in the correct position. The simulation confirms the expected behavior: the LED turns on in darkness and turns off in bright light.

Therefore, the correctness of the circuit is established by three forms of agreement:

Mathematical agreement

Wiring agreement

Simulation agreement

6.6.5 What is the final conclusion of the project?

The final conclusion of the project is that a small light-dependent voltage from an LDR voltage divider can safely control a larger LED current using a 2N2222 NPN transistor. The LDR and fixed resistor produce the control voltage at the SENSE node. The base resistor limits the transistor base current. The transistor acts as the switching element. The LED resistor limits the output current. In darkness, the SENSE voltage rises to approximately 4.55 V, the base current becomes approximately 0.819 mA, the transistor turns on, and the LED lights. In bright light, the SENSE voltage falls to approximately 0.45 V, the base current becomes approximately zero, the transistor turns off, and the LED remains off.

This confirms that the circuit successfully demonstrates the use of a transistor as an electronic switch controlled by a light-dependent sensor signal. The design is complete because the input signal is defined, the switching condition is calculated, the wiring is specified, the simulation behavior is predictable, and the main faults can be diagnosed through systematic measurement.